MIL-L-12997E
4 June 1976
SUPERSEDING
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18 September 1972

MILITARY SPECIFICATION

LINK, CARTRIDGE: METALLIC BELT, 20MM, M12

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of metallic belt link (welded or non-welded) intended to hold 20mm ammunition in the form of flexible belts for use in the M39 type 20mm automatic gun.

2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-116	-	Preservation-Packaging, Methods of
MIL-W-6858	-	Welding, Resistance: Aluminum, Magnesium, Non-Hardening Steels or Alloys, Nickel/Alloys, Heat-Resisting Alloys, and Titanium Alloys: Spot and Seam
MIL-W-13855	-	Weapons: Small Arms and Aircraft Armament Subsystems, General Specification for
MIL-P-16232	-	Phosphate Coatings, Heavy, Manganese or Zinc Base (for Ferrous Metals)
MIL-W-45223	-	Welding, Spot, Hardenable Steels
MIL-I-45607	-	Inspection Equipment, Acquisition, Maintenance and Disposition of

FSC 1305

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STANDARDS

Federal
Fed. Std. No. 356 - Commercial Packaging of Supplies and Equipment

Military
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1167 - Ammunition Data Cards
MIL-STD-1168 - Ammunition Lot Numbering

DRAWINGS

Rock Island Arsenal
D7147028 - Link, Cartridge, Metallic Belt, 20mm, M12

(Product drawings referenced in this specification form a part of D7147028.)

PUBLICATIONS

Rock Island Arsenal
P7147028 - Packaging Data Sheet for Link, Cartridge: Metallic Belt, 20mm, M12
QE-G-DL7147028 - Quality Engineering Data List - Link, Metallic, 20mm, M12

(Acceptance Inspection Drawings referenced in this specification form a part of QE-G-DL7147028.)

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM)
ASTM E18 - Test for Rockwell Hardness of Metallic Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA, 19103.)

3. REQUIREMENTS

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3.1 First article. Requirements for submission of first article shall be as specified in the contract (see 6.1). Unless otherwise specified (see 6.1), the first article shall include the pilot pack (see 5.1).

3.2 Materials, construction and design. The links shall conform to the materials, construction and design requirements specified herein, on Drawing D7147028 and in MIL-W-13855.

3.2.1 Pull. Links shall be capable of withstanding 600 + 25 pounds tensile load without fracture. When tensile load is applied, the distance X-X as shown on Drawing D7147028 shall not exceed 1.945 inches at load and the bridge shall be tight on the body after test. Testing shall be as specified in 4.5.3.5 or 4.5.3.5.1, as applicable.

3.2.2 Twist. Links assembled into belts of 25 links with Government standard 20mm dummy cartridges shall be capable of withstanding a 180° twist with no evidence of partial or complete belt separation. Testing shall be as specified in 4.5.3.6.

3.2.3 Cartridge retention. The force required to strip the Government standard 20mm dummy cartridges from links shall be not less than 125 pounds nor more than 300 pounds when applied parallel to the longitudinal centerline of the cartridge. Testing shall be as specified in 4.5.3.7.

3.2.4 Free hinging. Links assembled into belts of 25 links with Government standard 20mm dummy cartridges shall hinge and fold over without binding, kinking or any interference between the projecting surfaces of the links. Testing shall be as specified in 4.5.3.8.

3.2.5 Welding. Welding of the stress bridge shall be in accordance with MIL-W-45223 or Class B of MIL-W-6858. Testing shall be as specified in 4.5.3.9.

3.2.6 Functioning. Links shall be capable of being assembled with Government standard 20mm cartridges into belts and shall operate in the normal firing of an M39 type automatic gun without malfunctioning, cracking, breaking or separating. Testing shall be as specified in 4.4.

* 3.3 Marking. Links shall be marked in accordance with MIL-W-13855 and with the link model number as specified on Drawing F7147027.

* 3.3.1 Lot identification. Links shall be identified with the inspection lot number as specified in MIL-STD-1168.

* 3.3.2 Data cards. Data cards conforming to MIL-STD-1167 shall be prepared for each lot of links and shall include the information specified in the contract (see 6.1).

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3.4 Workmanship. Workmanship shall be in accordance with MIL-W-13855 with the exception that cracks, tears, burs and deformations shall not exceed the limits defined in 6.2.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality assurance terms and definitions. Quality assurance terms and definitions used herein are in accordance with MIL-STD-109.

4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. The first article (initial production unit(s)) shall be submitted for inspection in accordance with the contract (see 6.1). The first article shall be representative of the production processes to be used during quantity production. The first article shall be subjected to the quality conformance inspection specified herein, the functioning test specified in 4.6.10 and such other inspection as necessary to determine that all requirements of the contract have been met.

* 4.4.1 Examination and test. Five of each component will be inspected for all unlisted drawing dimensions and tests.

4.5 Quality conformance inspection.

4.5.1 Inspection lot. Unless otherwise specified herein, the formation, size and presentation of inspection lots shall be in accordance with MIL-STD-105 and MIL-W-13855. Samples shall be withdrawn in accordance with MIL-STD-105 from packaged links. Inspection lots shall not exceed 500,000 links per lot.

* 4.5.1.1 Heat treat lot. The heat treat lot shall be limited to one heat of steel and shall consist of assemblies or components heat treated in a batch type furnace at one charge or in a continuous type furnace during an uninterrupted period extending over not more than one eight hour shift.

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4.5.2 Examination. Examination of links and packaging shall be performed as specified herein and in MIL-W-13855.

4.5.2.1 Classification of defects. Classification of defects and acceptable quality levels (AQL's) for examination in accordance with MIL-STD-105 shall be as specified herein and shall include the following provisions:

a. Unless otherwise specified, AQL's apply to each listed characteristic.
b. Where a Government drawing is listed as the method of inspection, the contractor shall use inspection equipment fabricated in accordance with the applicable drawings. In the event the contractor desires to use alternate equipment which equals or exceeds the accuracy requirements of the specified drawings, it shall be submitted to the Government for approval.
* c. Where Visual is specified as the method of inspection of protective coatings, the coating shall be visually examined for completeness, uniformity in appearance and color, for freedom from pits, corrosion, scratches, and worn or bare spots.
* d. When Visual is specified as the method of dimensional inspection, the sample shall be either scaled or compared with a specimen of known acceptable quality.
* e. Inspection for Workmanship shall include examination for missing manufacturing operations and the presence of nicks, cracks, scratches, tool marks, deformations or other signs of poor workmanship.
* f. Where SME (Standard Measuring Equipment) is specified as the method of inspection, the contractor may use any type of industry developed, commercially available, multiusage equipment or special inspection equipment approved by the government.

4.5.2.1.1 Link Cartridge, Metallic Belt, 20mm, M12 (Drawing D7147028).

Categories	Characteristics	AQL	Inspection Method
Critical:	None defined		
Major:			
* 101	Cracks and tears in principle areas and burs as defined in 6.2 (Cracks and tears in non-principle areas and deformations are covered under minor 210-Workmanship)	0.065	Visual
102	Missing or improper weld buttons (for welded link only)	0.065	Visual
* 103	Width; tip flare, stress arm (w/o ring)	0.065	A8415842, A8415843

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Categories	Characteristics	AQL	Inspection Method
Major:			
104	Max. clearance between bridge or body arms	0.25	A7190493
* 105	Min. distance from bridge arms or body arms to vertical center line of link (w/steel plug B7190405 in link)	0.25	SME
* 106	Min. distance from tip of flared end of body stress arm (w/ring) to longitudinal center line of link (w/steel plug B7190405 in link)	0.25	SME
107	Grip on cartridge (min. opening)	0.65	C7190032
108	Height of bridge tip from bridge base	0.65	C7190492
109	Width; tip of stress bridge (for non-welded link only)	0.65	A8415859
110	Guide tabs offset height; inner surface max., and outer surface min. (w/steel plug B7190405 in link before retention test)	0.65	B7190256
111	Width; flared end stress arm (w/ring)	1.0	Visual
112	Missing or improper detents	1.5	Visual
Minor:			
201	Missing or defective protective coating	2.5	Visual
202	Missing or improper marking (see 3.3)	2.5	Visual
203	Overall length; guide tabs	2.5	B7190249
204	Overall length of link	2.5	B7190053
205	Inside diameter of connector ring	2.5	A7190052
206	Improper clearance; feeder channel guide	2.5	D7190267
207	Location: front end of front guide tab (w/steel plug B7190405 in link)	2.5	B7190252
208	Max. overall length; longitudinal offset of guide tabs (w/steel plug B7190405 in link)	2.5	B7190252
209	Min. distance; front surface of ramming guide embossment to rear surface or rear guide tab longitudinal offset	2.5	A8415858
* 210	Workmanship (includes cracks and tears in non-principle areas and deformations as covered in 6.2)	4.0	Visual

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4.5.2.1.2 Packaging. Unless otherwise specified in each listed defect, the packaging requirements are specified on Packaging Data Sheet P7147028. The inspection method shall be visual.

Categories	Characteristics	AQL
Major:		
101	Improper level of preservation and packing (see procurement documents)	1.0
102	Incorrect or illegible marking	1.0
103	Inadequate cleaning and drying	1.5
* 104	Improper preservative oil application and drainage (Improper drainage is the presence or accumulation of liquid oil which is not held as an integral part of the phosphate coating and is evident as droplets, puddles or as an oil film so wet as to be moving visually.)	1.5
105	Improper containers	1.5
106	Container interior contaminated with dirt, lint, or other foreign matter	1.5
107	Improperly oriented links in the unit pack.	1.5
108	Missing volatile corrosion inhibitor treated sheets between layers of links (when applicable)	1.5
109	Incorrect quantity of links in shipping containers	1.5
110	Improper closure of shipping container	1.5
Minor:		
201	Workmanship	4.0

4.5.3 Testing.

4.5.3.1 Certification. Unless otherwise specified, the contractor shall furnish the Government representative with certified statements that each inspection lot conforms to the materials and processes specified on the applicable drawings and specifications.

* 4.5.3.2 Accelerated corrosion testing. Sample links shall be selected from inspection lots for each processing bath prior to application of supplementary oil treatment. Lot formation and sampling shall be in accordance with MIL-P-16232 whether manual basket dip or continuous conveyor equipment is used. The sample links shall be tested for compliance with the requirements of MIL-P-16232 using the test methods specified in 4.6.1.

* 4.5.3.3 Hardness testing. Sample assemblies or components shall be taken from each heat treat lot in accordance with MIL-STD-105, using a

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single sampling plan, normal inspection, Inspection Level S-4 and tested for hardness using the test method specified in 4.6.2. Acceptance shall be based on an AQL of 0.40 percent defective.

* 4.5.3.4 Decarburization testing. A sample of 5 links shall be selected at random from each heat treat lot and tested for decarburization using the test method specified in 4.6.3. Failure of any link shall be cause for rejection of the heat treat lot.

* 4.5.3.5 Pull testing. A sample of 32 links shall be taken from each heat treat lot and tested for pull (see 3.2.1) using the test method specified in 4.6.4. Failure of any link shall be cause for rejection of the heat treat lot.

* 4.5.3.5.1 Pull testing (alternative). When the alternative method of construction is used, sample links shall be taken from each inspection lot in accordance with MIL-STD-105, using a single sampling plan, normal inspection, Inspection Level I. Testing for pull (see 3.2.1) shall be performed using the test method specified in 4.6.4. Acceptance shall be based on an AQL of 0.40 percent defective.

4.5.3.6 Twist testing. A separate sample of 800 links from each inspection lot shall be tested for twist (see 3.2.2) using the test method specified in 4.6.5. Failure of any link to meet the requirement shall be cause for rejection of the lot.

* 4.5.3.7 Cartridge retention testing. A sample of 80 links from each inspection lot shall be tested for cartridge retention (see 3.2.3) using the test method specified in 4.6.6. Failure of two or more links shall be cause for rejection of the inspection lot.

4.5.3.8 Free hinging testing. The 800 link sample used in 4.5.3.6 shall be tested for free hinging (see 3.2.4) using the test method specified in 4.6.7. Failure of any link to meet the requirement shall be cause for rejection of the lot.

4.5.3.9 Welding testing. Welding (see 3.2.5) shall be tested using the test method and sampling procedures specified in 4.6.8.

4.5.3.10 Determination of cleanliness testing. Links shall be tested for cleanliness using the sampling procedure and test methods specified in 4.6.9.

4.5.4 Inspection equipment.

* 4.5.4.1 Acquisition, maintenance and disposition. Unless otherwise specified (see 6.1) responsibility for acquisition, maintenance and

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disposition of acceptance inspection and test equipment prescribed on the Quality Engineering Data List QE-G-DL7147028 and for all other inspection equipment required by applicable specifications shall be in accordance with MIL-I-45607.

4.5.4.2 Accuracy of standard measuring equipment. When commercial and modified commercial inspection and test equipment is used, it must be capable of repetitive measurements to an accuracy of 10 percent of the total tolerance of the characteristic being inspected.

4.6 Test methods.

* 4.6.1 Accelerated corrosion test. Sample links shall be tested for corrosion resistance of the applied protective coating using either salt spray (fog) test or Springfield Immersion Test specified in MIL-P-16232.

* 4.6.2 Hardness test. The sample assemblies or components shall be tested for hardness requirements using ASTM E18.

* 4.6.3 Decarburization test. The cross section of links shall be polished, etched with a 2-5 percent nital solution and examined under a microscope of not less than 500 power to determine compliance with decarburization requirements of Drawing D7147028.

4.6.4 Pull test. Each sample link shall be tested using the inspection equipment specified on Drawing F7190358. The links shall be inserted in the fixture and subjected to a 600 pound load.

4.6.5 Twist test. The sample links shall be tested by assembling the links with Government standard 20mm dummy cartridges into belts of 25 linked cartridges. Each belt shall be laid out horizontally to its full length with the open side of the links down, and one end of the belt shall be flipped sharply by hand 180° to its other side so that the twisting action will progressively move along the belt to the free end. The test shall be repeated with the belt positioned with the open side of the links up.

4.6.6 Cartridge retention test. The sample links shall be tested by using the inspection equipment and method specified on Drawing F7190403. Partial extraction shall be classified as a failure (see 6.2.5).

4.6.7 Free hinging test. The sample links shall be tested by assembling the links with Government standard 20mm dummy cartridges into belts of 25 linked cartridges. Each belt shall be laid out horizontally to its full length, with the open side of the links up, and one end shall be drawn back upon the remainder of the belt until the belt is completely reversed (open side of links down). One end of the reversed belt shall then be drawn back upon the remainder of the belt until the belt is returned to its original (open side of links up) position.

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* 4.6.8 Welding test. The links shall be tested for welding requirements using MIL-W-45223 or MIL-W-6858.

* 4.6.9 Determination of cleanliness test. The links shall be tested for cleanliness requirements in accordance with MIL-P-116.

4.6.10 Functioning test. A sample of 200 links taken from first article shall be tested for the functioning requirement (see 3.2.6) by assembling the links into belts using Government standard 20mm cartridges and firing the belts in bursts of 50 rounds each in an M39 type automatic gun.

5. PACKAGING

5.1 Pilot pack. A pilot pack shall consist of links preserved in accordance with Packaging Data Sheet P7147028 for the level of protection specified in the contract (see 6.1), packed level B and forwarded as specified in the contract (see 3.1).

5.2 Levels A and B. Preservation, packing and marking shall be in accordance with Packaging Data Sheet P7147028 for the level of protection specified in the contract (see 6.1).

5.3 Commercial packaging. Preservation, packing and marking shall be in accordance with Fed. Std. No. 356.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements:

a. Title, number and date of this specification.
b. List of drawings, specifications and publications pertinent to the link showing applicable revision dates.
* c. Data card information required (see 3.3.2).
d. Requirements for submission of first article and pilot pack (see 3.1, 4.4 and 5.1).
e. Responsibilities for acquisition, maintenance and disposition of acceptance inspection equipment, if other than specified (see 4.5.4.1).
f. Responsibility for furnishing dummy cartridges.
g. Selection of applicable level of protection (A or B), or commercial packaging (5.1, 5.2 and 5.3).

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6.1.2 Contract data requirements. When warranted, requirements for reports of the results of examination and testing should be specified on a DD Form 1423 included in the contract.

6.1.3 When warranted, the contract should specify the application of MIL-I-45208 or MIL-Q-9858 on the Management Control Systems Summary List, DD Form 1660.

6.1.4 Unless otherwise specified (see 6.1), the contract should specify the application of MIL-I-45607 on the Management Control Systems Summary List, DD Form 1660.

6.2 Definition of terms.

6.2.1 Deformed links. Links whose shape or dimensions, or both, have been mutilated or altered by processing to the extent that the links are unsuitable for their intended use.

6.2.2 Cracks. See figures 1, 2, and 3.

6.2.2.1 Cracks in principal areas. Any visible fracture of the link extending through the stock thickness in depth or a distance in length equal to or greater than the material stock thickness.

6.2.2.2 Cracks in nonprincipal areas. Any visible fracture of the link with ragged or irregular edges extending through the stock thickness in depth and equal to or greater than the stock thickness in length; or any visible fracture less than stock thickness in depth and greater than two times stock thickness in length.

6.2.3 Burs. Any rough edge on any part of the link, which in the opinion of the Government representative, would produce a definite removal of metal from a cartridge case, cause injury to personnel in handling, or result in binding of the link.

6.2.4 Principal areas. The body arms (except for the edges facing inward between the flared tips), stress bridge arms, and the connecting ring.

6.2.5 Partial extraction. When only one end of the mandrel or round becomes free.

6.3 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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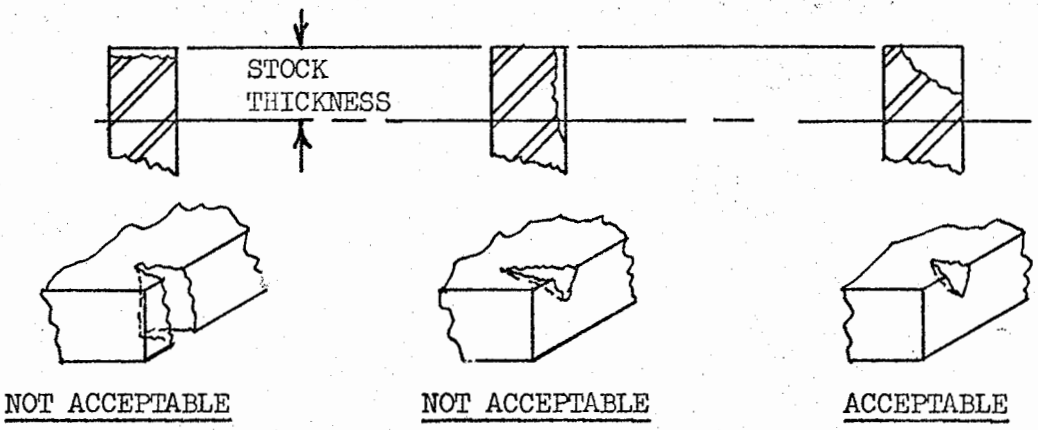
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Air Force - 70

Reviewer:
Army - MU

Preparing Activity:
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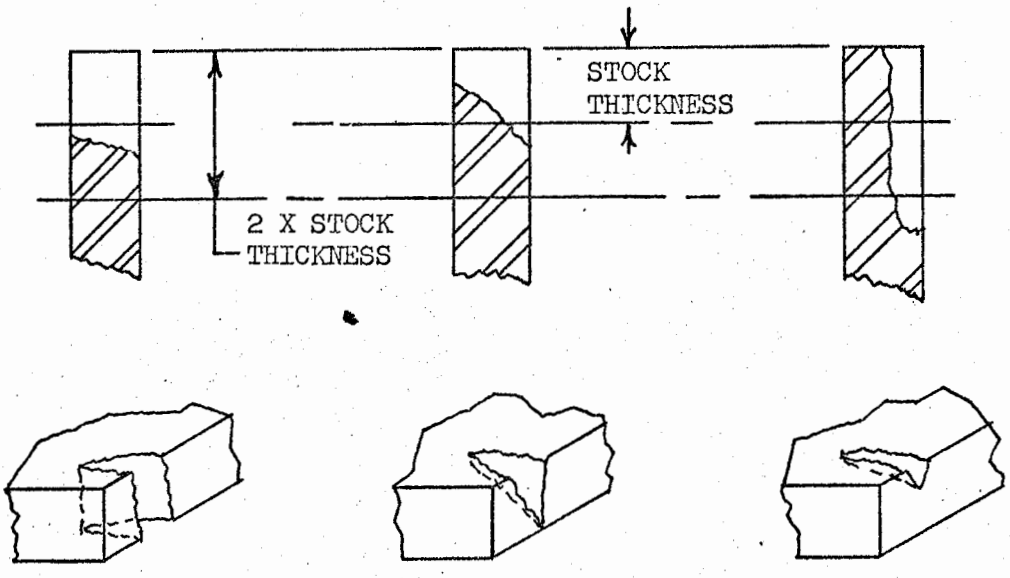
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CRACKS IN PRINCIPAL AREAS

Figure 1

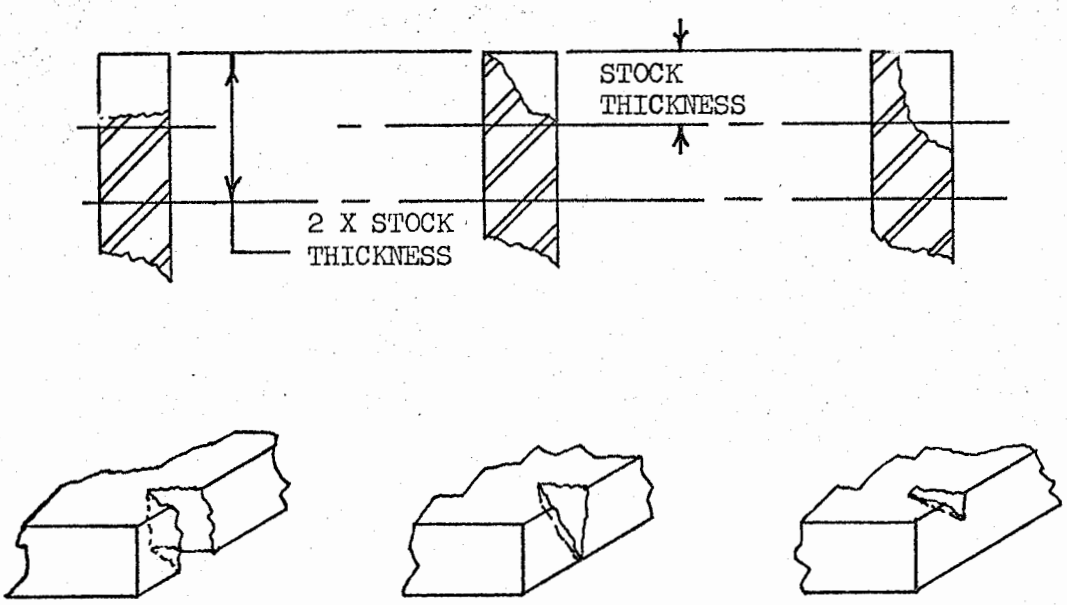
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ALL NOT ACCEPTABLE

CRACKS IN NONPRINCIPAL AREAS

Figure 2



ALL ACCEPTABLE

CRACKS IN NON PRINCIPAL AREAS

Figure 3

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